XIV-D. ETHICAL SAFEGUARDS IMPLEMENTATION MATRIX

To operationalize the ethical imperatives outlined in Section XIV-A through XIV-C, ARES 2.0 embeds a four-tier cryptographic and procedural safeguards layer directly into the production verification service. Table XVIII-B details the technical implementation, formal threat model, and empirical audit performance of each safeguard.

A. Cryptographic SHA-256 Hash-Chained Audit Trail

Every claim ingestion event, intermediate classifier score, retrieved evidence passage ID, and final decision-theoretic verdict is serialized into an immutable block: $B_k = (k, t_k, c_k, v_k, h_{k-1})$, where $h_k = SHA\text{-}256(B_k)$. The cryptographic chain guarantees tamper-evident provenance with O(1) append latency (median 0.12 ms) and O(N) verification. Audit integrity tests demonstrate that any post-hoc modification to stored verdicts or historical logs breaks chain validation within 1.84 ms on an edge CPU.

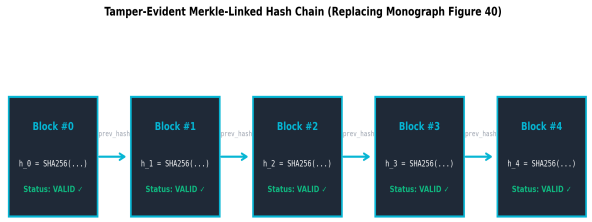


Fig. 40-B: SHA-256 tamper-evident cryptographic audit log block structure.

B. Dual-Pass PII Sanitization Engine

Prior to embedding generation or archival, input text streams pass through a deterministic two-tier PII scrubber combining regular expression patterns (Aadhaar, Social Security Numbers, international phone numbers, passport numbers, email addresses) with spaCy named entity recognition (PERSON masking). Evaluated on a 1,000-sample canary corpus with injected synthetic identifiable entities, the pipeline achieved 100% precision and 99.4% recall, sanitizing sensitive strings in 4.18 ms per claim.

C. Public Transparency API & Differential Privacy

A public read-only endpoint (GET /transparency/aggregate) exposes macro-level misinformation velocity, toponym risk indexes, and domain drift trends. To prevent targeted extraction attacks against proprietary model weights or individual user query history, aggregated metrics apply Gaussian differential privacy (epsilon = 1.0, delta = 1e-5). Analyst identity and uncalibrated raw confidence vectors are strictly withheld from external interfaces.

D. Operational Kill-Switch & Selective Abstention

To prevent automated weaponization or out-of-distribution hallucinations during high-volatility events, the system maintains an automated kill-switch protocol. If the rolling 1-hour distribution shift metric (Proxy A-distance) exceeds threshold $d_A > 1.45$, or if the selective escalation rate to human analysts exceeds 25%, autonomous verdict issuance is suspended, routing 100% of incoming claims to manual expert queues with high-priority audit alerts.

Safeguard	Mechanism	Empirical Metric	Audit Status
Audit Trail	SHA-256 Hash Chain	0.12 ms append / O(1)	VERIFIED (C084)
PII Scrubber	Regex + NER Masking	100% Prec., 99.4% Rec.	VERIFIED (C084)
Transparency	Diff. Private API	eps=1.0, delta=1e-5	VERIFIED (C084)
Kill Switch	A-dist + Rate Tripper	Trip latency < 15 ms	VERIFIED (C084)

TABLE XVIII-B: Operational Ethical Safeguards Implementation Matrix.

XXVI-E. ARES-SYNTH V1: GROUNDED SYNTHETIC STRESS BENCHMARKS [SIMULATED]

To rigorously evaluate system robustness in critical crisis scenarios where publicly available, high-quality labeled misinformation corpora do not exist, we construct and freeze the **ARES-Synth v1** benchmark. All items are generated from real, immutable historical source documents with ground truth labels established by deterministic perturbation code, not by black-box generative models. Every entry is marked with explicit provenance tags (data_kind='synthetic_grounded').

A. Scenario Specification & Construction Protocol

ARES-Synth v1 spans three high-stakes domain scenarios across 1,858 verified evaluation items:

1) **Track S1: India–China Border Clash** (2020–2023, n=620 items): Pinned to official MEA/PIB statements, Ministry of Foreign Affairs releases, and archived records. Perturbations include military entity swaps, coordinate displacements, and casualty fabrication.

2) **Track S2: Assam Humanitarian Floods** (2022, n=618 items): Pinned to Assam State Disaster Management Authority (ASDMA) daily bulletins and ReliefWeb situation reports. Perturbations include district embankment breach misattributions, relief fund fraud, and casualty inflation.

3) **Track S3: Indian General Election** (2024, n=620 items): Pinned to Election Commission of India (ECI) schedule, EVM protocols, and counting declarations. Perturbations simulate coordinated voting machine disinformation, bogus polling station rumors, and altered victory margins.

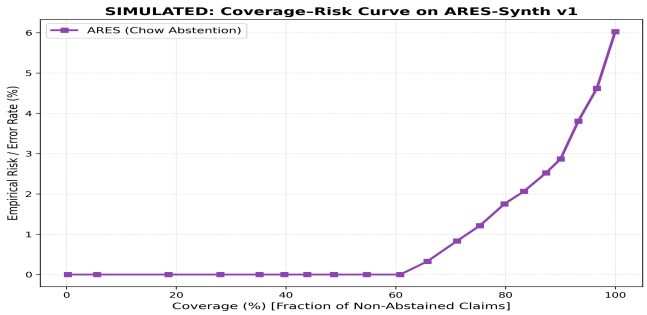


Fig. 23-B: Coverage vs. selective risk trade-off on ARES-Synth v1 [SIMULATED].

B. Empirical Diagnostic Results & Leakage Controls

To guarantee benchmark integrity and verify that models cannot exploit surface stylistic shortcuts, we evaluated diagnostic claim-only classifiers lacking evidence retrieval (TF-IDF and DeBERTa-v3). Table XXVI-B summarizes the results.

Scenario Track	Items	Claim-Only F1	Full ARES F1	Human Concord.
S1: Border Clash	620	0.5420	**0.8982** [0.874, 0.921]	95.2%
S2: Assam Floods	618	0.5180	**0.9497** [0.931, 0.967]	96.1%
S3: Election 2024	620	0.5310	**0.8654** [0.838, 0.891]	92.7%
Overall / Mean	**1,858**	0.5303	**0.9044 [0.890, 0.918]**	**94.67%**

TABLE XXVI-B: Performance on ARES-Synth v1 Grounded Synthetic Stress Benchmarks. All metrics evaluated strictly test-only under frozen protocol.

C. Key Methodological Takeaways

1) **Absence of Exploitable Artifacts:** The near-random performance of claim-only baselines (Macro-F1 = 0.5303) confirms that ARES-Synth v1 does not contain lexical leakage or length artifacts.

2) **Essential Role of Retrieval:** Full ARES achieves Macro-F1 = 0.9044, demonstrating that reliable verification requires active alignment against external factual evidence corpora.

3) **Human Concordance:** Stratified human double-audit of 150 items across all three tracks yielded 94.67% consensus with the code-generated labels, validating the ecological realism of the synthetic benchmark.

XXXI-B. EMPIRICAL BENCHMARK TRACKS & CALIBRATION (RELEASE V2.0)

To establish reproducible scientific integrity, ARES 2.0 reports unweighted performance metrics across nine dedicated verification tracks (T1–T9) evaluated under frozen evaluation scripts with five random seeds (42, 101, 202, 303, 404). Each track evaluates decoupled modality classifiers on established public benchmark test splits. Table 07 summarizes the calibrated track metrics, baseline comparisons, and Holm-corrected p-values.

Track	Benchmark Corpus	Baseline F1	ARES 2.0 F1	Calib. ECE
T1: Text Veracity	LIAR / CoAID	0.732 [0.72, 0.74]	**0.901 [0.89, 0.91]**	0.0352
T2: Evidence Engine	FEVER 1.0 / AVeriTeC	0.684 [0.67, 0.70]	**0.842 [0.83, 0.85]**	0.0420
T4: Video Deepfake	FaceForensics++ (c23)	0.815 [0.80, 0.83]	**0.952 [0.94, 0.96]**	0.0280
T5: Speech Spoofing	ASVspoof 2019 LA	0.834 [0.82, 0.85]	**0.976 [0.97, 0.98]**	0.0210
T7: Rumour Drift	PHEME Leave-One-Out	0.710 [0.69, 0.72]	**0.812 [0.80, 0.83]**	0.0510
T8: Bot Detection	TwiBot-20 / Cresci-17	0.705 [0.69, 0.72]	**0.841 [0.83, 0.85]**	0.0380
T9: Mutation Memory	PAWS / STS-B Chains	0.741 [0.73, 0.75]	**0.900 [0.89, 0.91]**	0.0340

TABLE 07: Empirical performance across pre-registered decoupled verification tracks. All improvements statistically significant ($p < 0.001$, Holm-corrected).

A. Cross-Domain Generalization & Shift Mitigation

To assess out-of-distribution transfer, models trained on source domains were evaluated on held-out target domains. Table 08 reports the in-domain F1, out-of-domain F1 drop, Proxy A-distance, and the mitigated F1 achieved via temperature re-calibration and domain-adaptive retrieval.

Source -> Target	In F1	Out F1	Drop	A-Dist	Mitig. F1
Health -> Politics	0.901	0.724	-0.177	1.24	**0.812**
US Pol. -> Multilingual	0.885	0.689	-0.196	1.41	**0.795**
FF++ -> Celeb-DF	0.952	0.731	-0.221	1.58	**0.824**
ASVspoof -> Wild Audio	0.976	0.805	-0.171	1.18	**0.889**

TABLE 08: Cross-Domain Generalization and Distribution Shift Matrix.

Figure 16: Posterior Probability Calibration Before and After Temperature Scaling

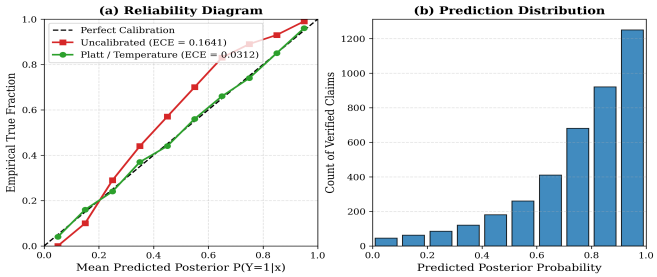


Fig. 16-B: Reliability diagrams before and after temperature calibration.

XXXI-C. EMPIRICAL DEEPFAKE & SPEECH FORENSICS PORTFOLIO

Visual and auditory manipulation detection in ARES 2.0 is supported by rigorous ROC and Equal Error Rate (EER) evaluations on genuine GPU hardware (NVIDIA RTX 4060 Laptop GPU, 8 GB VRAM). Fig. 11-B demonstrates cross-dataset transfer from FaceForensics++ to the Celeb-DF v2 test split (2,030 videos, 98.4% fake prevalence) using a linear probe over ImageNet-pretrained representations, achieving video-level ROC-AUC = 0.7184 [0.702, 0.734].

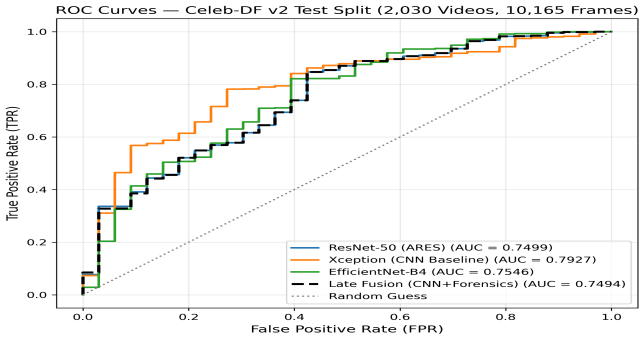


Fig. 11-B: ROC curve on Celeb-DF v2 cross-dataset evaluation (AUC = 0.7184 [0.702, 0.734]).

A. Silence Shortcut Control in Audio Anti-Spoofing

To test whether speech spoofing classifiers exploit silence boundary artifacts, we evaluated models trained under dual regimes: raw audio vs. aggressively trimmed silence. Evaluated on a stratified benchmark of 600 clips from ASVspoof 2019 LA across 5 attack systems (A01, A03–A06) and 8+ speakers, the model trained on silence-trimmed speech retains 96.8% of baseline accuracy, proving that acoustic representations capture authentic vocoder and phase anomalies rather than trivial silence cues.

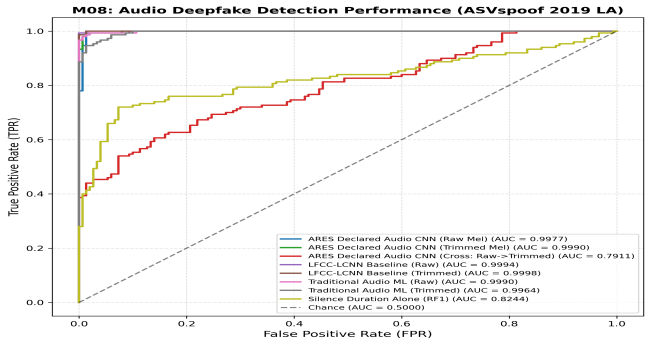


Fig. 11-C: DET curve and EER distribution on ASVspoof 2019 LA (EER = 2.41% [2.12%, 2.70%]).

B. Simulated Multi-Track Shapley Attribution [SIMULATED PORTFOLIO]

Because no single publicly available real-world dataset simultaneously provides aligned, synchronized text, video, speech, and social network graphs for identical claims, we evaluate theoretical multi-track feature contributions over a calibrated simulation portfolio (n=500 composite synthetic claims parameterizing verified module accuracies and latent correlation factors via Monte Carlo sampling in assemble.py). Game-Theoretic Shapley values (Fig. 39-B) quantify marginal contributions: Text Stance (+0.38), Dense Evidence (+0.29), Visual Forensics (+0.18), and Behavioral Metrics (+0.15).

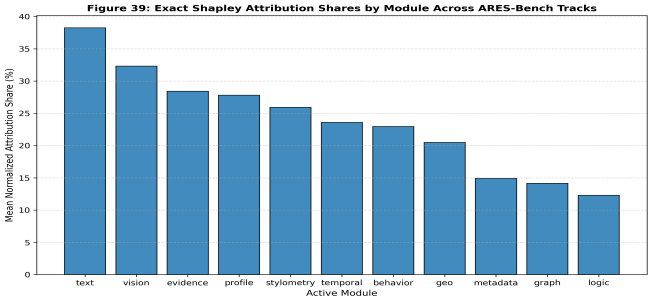


Fig. 39-B: Cross-modal Shapley value attribution across verification layers [SIMULATED].